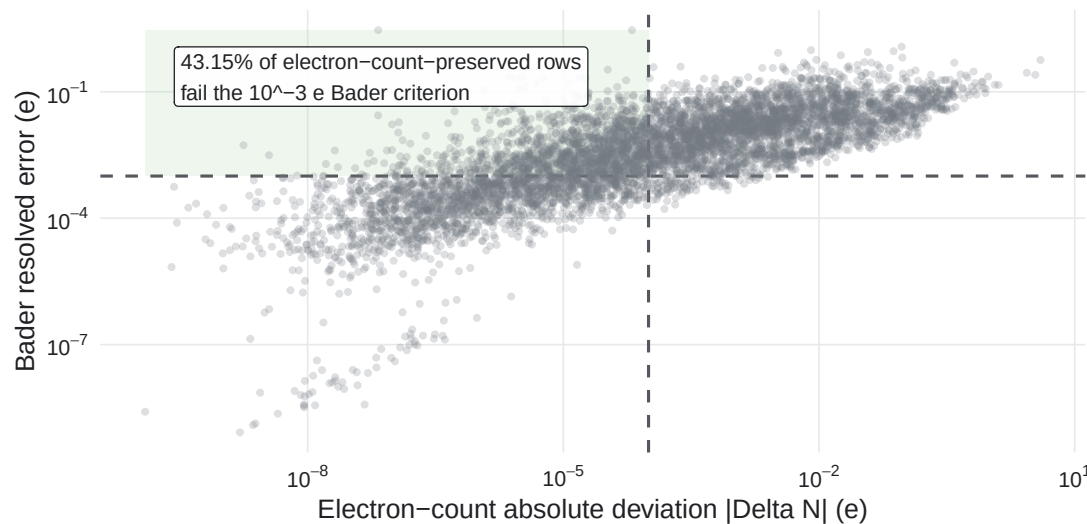


Figure 2 | QoI hierarchy reveals operator-dependent error propagation

global linear: electron count $\rightarrow$ smooth nonlocal: Hartree potential $\rightarrow$ topology-dependent: Bader charge

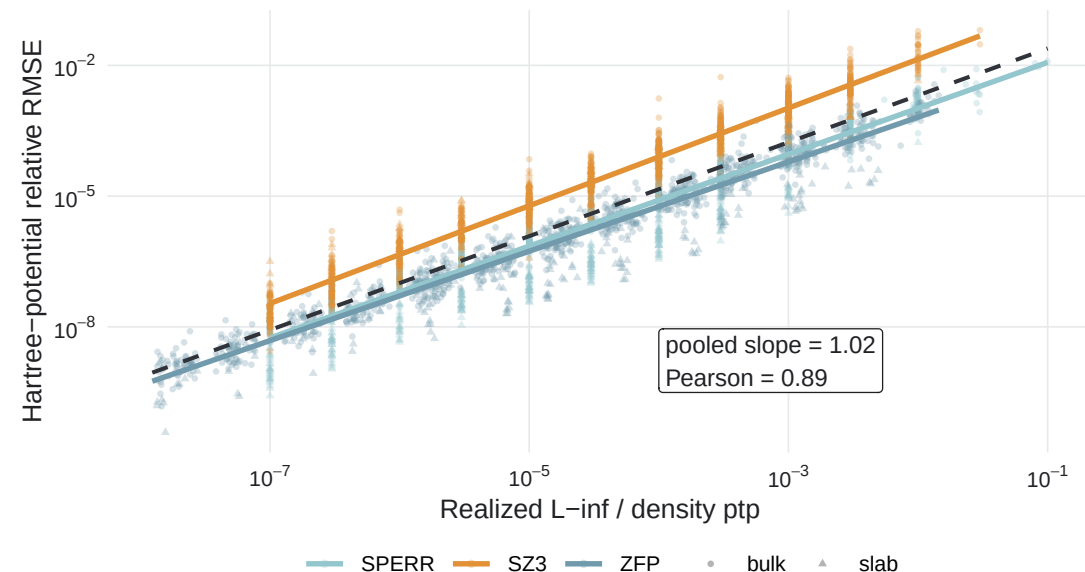
A | Global conservation does not guarantee local fidelity

Same reconstructions; dashed thresholds mark $|\Delta N| = 10^{-4}$ e and Bader error = 10^{-3} e



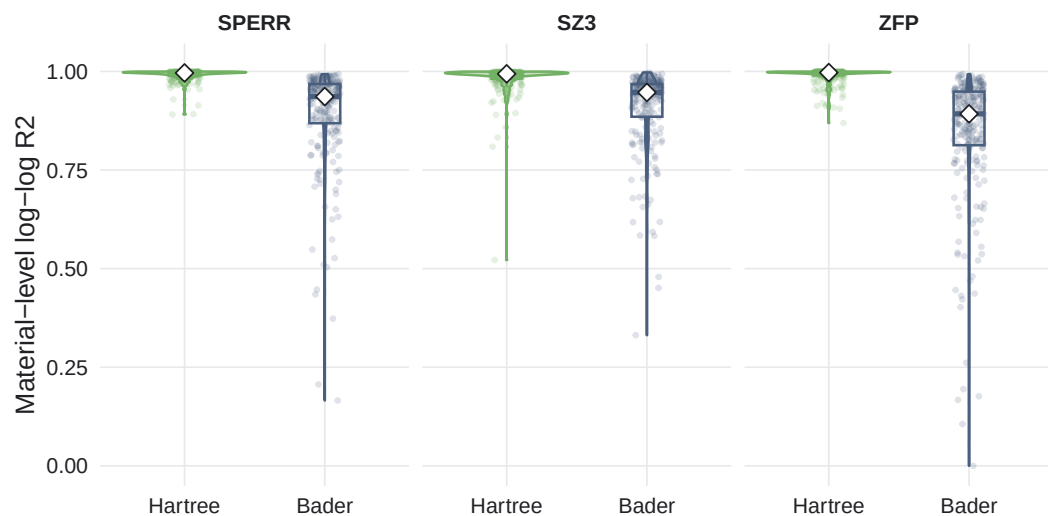
B | Hartree potential follows a smooth near-linear response

All codecs and bulk/slab systems; dashed line: pooled log-log fit



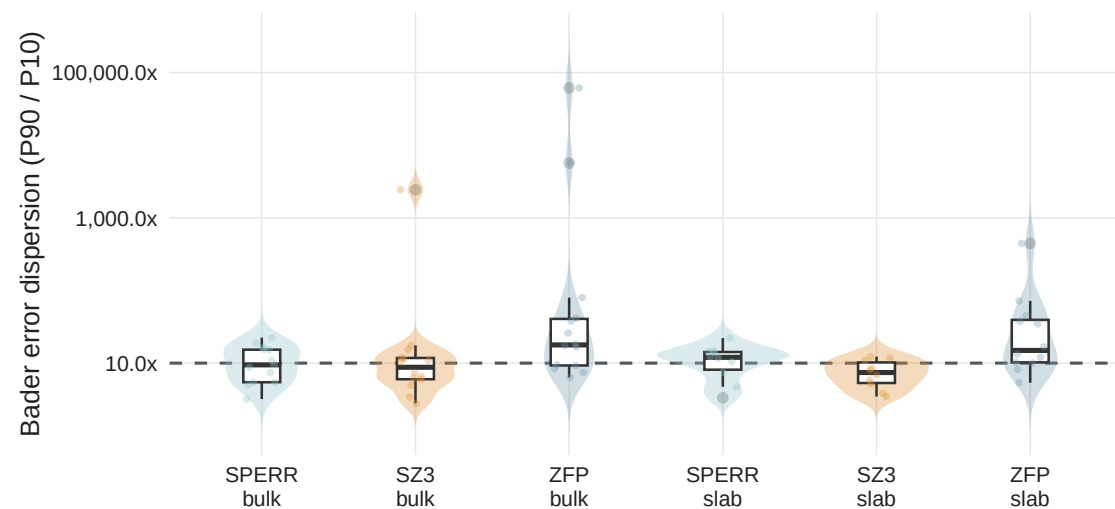
C | Operator structure changes error regularity

Material-level log-log R2; Hartree median 0.994–0.997, Bader median 0.89–0.95



D | Similar Hartree fidelity does not imply similar Bader fidelity

P90/P10 spread within 0.5-decade Hartree-error bins; 55.4% of rows lie in bins with at least 10x spread



Hartree statistics use rows that pass the pre-specified reconstruction and reproduction checks.

Bader irregularity is summarized by material-level smoothness and matched-Hartree dispersion rather than universal rung-wise monotonicity.